

The Irrationality of $\sqrt{2} + \sqrt{3}$

Formal status: Kernel verified

Faithfulness: User approved; independently reviewed by claude-opus-5 on claude (agreed)

Informal completeness: Not independently assessed

Document status: TeX compiled

Theorem

The real number $\sqrt{2} + \sqrt{3}$ is irrational; that is, there is no rational number q with $q = \sqrt{2} + \sqrt{3}$.

Formally (Lean 4 / Mathlib):

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theorem irrational_sqrt_two_add_sqrt_three : Irrational (Real.sqrt (2 : ℝ) + Real.sqrt (3 : ℝ))
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Here ‘Irrational x ’ is Mathlib’s definition ‘ $x \notin \text{Set.range } ((\uparrow) : \mathbb{Q} \rightarrow \mathbb{R})$ ’, i.e. x is not the image of any rational number under the canonical embedding $\mathbb{Q} \hookrightarrow \mathbb{R}$.

Approved interpretation

- $\mathbb{R}$ — the real numbers, Lean/Mathlib’s ‘Real’
- $\mathbb{Q}$ — the rational numbers, entering only through Mathlib’s ‘Irrational’ predicate, defined as ‘ $x \notin \text{Set.range } ((\uparrow) : \mathbb{Q} \rightarrow \mathbb{R})$ ’
- ‘ $\text{Real.sqrt} : \mathbb{R} \rightarrow \mathbb{R}$ ’ — the nonnegative square root function on the reals
- None explicit: the statement is a closed proposition about the single real number ‘ $\text{Real.sqrt } (2 : \mathbb{R}) + \text{Real.sqrt } (3 : \mathbb{R})$ ’.
- One universal quantifier is implicit inside ‘Irrational’: for every ‘ $q : \mathbb{Q}$ ’, ‘ $(q : \mathbb{R}) \neq \text{Real.sqrt } 2 + \text{Real.sqrt } 3$ ’.
- No hypotheses; the theorem is unconditional and has no binders.
- The numerals 2 and 3 are the real numbers 2 and 3, obtained via ‘OfNat $\mathbb{R}$ ’; they are ascribed as ‘ $(2 : \mathbb{R})$ ’ and ‘ $(3 : \mathbb{R})$ ’ so numeral elaboration is pinned rather than left to unification.
- ‘ Real.sqrt ’ is Mathlib’s total square-root function, which on nonnegative arguments returns the nonnegative root, so ‘ $\text{Real.sqrt } 2$ ’ and ‘ $\text{Real.sqrt } 3$ ’ are the intended principal roots.

- ‘Irrational’ and ‘Real.sqrt’ are available from Mathlib (‘Mathlib.Analysis.SpecialFunctions.Sqrt’ / ‘Mathlib.RingTheory.Int.Basic’ transitively via ‘import Mathlib’); ‘Irrational’ lives in the root namespace, so it is written unqualified.
- Repair applied: the binder field is now genuinely empty rather than a parenthetical English note, so the assembled signature is ‘theorem irrational_sqrt_two_add_sqrt_three : Irrational (Real.sqrt (2 : ℝ) + Real.sqrt (3 : ℝ))’ with nothing spurious between the name and the colon.
- Repair applied: explicit ‘(2 : ℝ)’ and ‘(3 : ℝ)’ type ascriptions on the numerals, and fully qualified ‘Real.sqrt’, to avoid any ambiguity in numeral or name elaboration.
- ‘sqrt(2)’ and ‘sqrt(3)’ are read as the principal (nonnegative) real square roots.
- ‘is irrational’ is formalized with Mathlib’s ‘Irrational’ predicate rather than an ad hoc ‘ $\forall p q : \mathbb{Z}, q \neq 0 \rightarrow \dots$ ’ statement; the two are equivalent over $\mathbb{R}$.
- The claim concerns the single sum ‘Real.sqrt 2 + Real.sqrt 3’ (≈ 3.146), not the irrationality of the two summands separately, and asserts nothing about algebraic degree or minimal polynomials.
- No strengthening or weakening: the logical content is exactly the original claim.

Human-readable proof

Overview

The argument is the classical one. A sum of two square roots is hard to attack directly, but squaring it collapses the two roots into a single one: $(\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6}$. Rationality is preserved by squaring and by the field operations, so if $\sqrt{2} + \sqrt{3}$ were rational then $\sqrt{6}$ would be rational too. Since 6 is not a perfect square, $\sqrt{6}$ is irrational, and the contradiction is complete.

Preliminary: $\sqrt{6}$ is irrational

We use the standard fact that for a natural number n , the real number $\sqrt{n}$ is rational only when n is a perfect square. For $n = 6$ this gives irrationality directly, since $2^2 = 4 < 6 < 9 = 3^2$, so 6 lies strictly between consecutive squares and is therefore not a square.

For completeness, here is the self-contained descent argument for this particular case. Suppose $\sqrt{6} = a/b$ with integers a, b , $b > 0$, and with a/b in lowest terms. Then $a^2 = 6b^2$. The right-hand side is even, so a^2 is even, hence a is even; write $a = 2a'$. Substituting gives $4a'^2 = 6b^2$, i.e. $2a'^2 = 3b^2$. Now the left-hand side is even, so $3b^2$ is even, and since 3 is odd, b^2 is even, hence b is even. So 2 divides both a and b , contradicting the assumption that a/b was in lowest terms. Therefore $\sqrt{6}$ is irrational.

In the formal development this step is discharged by Mathlib’s ‘norm_num’ extension for irrationality of square roots (‘Mathlib.Tactic.NormNum.Irrational’), which certifies ‘Irrational ($\sqrt{6}$)’ by exhibiting 6 as strictly between the consecutive squares 2^2 and 3^2 .

The main argument

Suppose, for contradiction, that $\sqrt{2} + \sqrt{3}$ is *not* irrational. Unfolding the definition, this means there exists a rational number q whose image in $\mathbb{R}$ satisfies

$$(q : \mathbb{R}) = \sqrt{2} + \sqrt{3}.$$

We record three elementary facts about the square-root function on the nonnegative reals:

1. $(\sqrt{2})^2 = 2$, since $2 \geq 0$; 2. $(\sqrt{3})^2 = 3$, since $3 \geq 0$; 3. $\sqrt{2} \cdot \sqrt{3} = \sqrt{2 \cdot 3} = \sqrt{6}$, using multiplicativity of $\sqrt{}$ on nonnegative arguments.

Now square the supposed identity. Expanding the binomial and applying (1)–(3):

$$(q : \mathbb{R})^2 = (\sqrt{2} + \sqrt{3})^2 = (\sqrt{2})^2 + 2 \cdot \sqrt{2} \cdot \sqrt{3} + (\sqrt{3})^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}.$$

Rearranging this real-number identity for $\sqrt{6}$ gives

$$\sqrt{6} = ((q : \mathbb{R})^2 - 5) / 2.$$

The crucial point is that the right-hand side is the image of a *rational* number. Indeed, $\mathbb{Q}$ is a field and the embedding $\mathbb{Q} \hookrightarrow \mathbb{R}$ is a ring homomorphism, so the rational number $r := (q^2 - 5)/2$ — formed from q by squaring, subtracting 5, and dividing by 2, all inside $\mathbb{Q}$ — satisfies $(r : \mathbb{R}) = ((q : \mathbb{R})^2 - 5)/2 = \sqrt{6}$. Hence $\sqrt{6}$ lies in the range of the embedding $\mathbb{Q} \hookrightarrow \mathbb{R}$, i.e. $\sqrt{6}$ is rational.

This contradicts the preliminary result that $\sqrt{6}$ is irrational. Therefore no such rational q exists, and $\sqrt{2} + \sqrt{3}$ is irrational. ■

Remarks

- The same argument shows more generally that $\sqrt{m} + \sqrt{n}$ is irrational whenever mn is not a perfect square: squaring gives $m + n + 2\sqrt{mn}$, so rationality of $\sqrt{m} + \sqrt{n}$ would force rationality of $\sqrt{mn}$. - The hypothesis that **mn** is a non-square is what does the work, not that m and n individually are non-squares. For instance $\sqrt{2} + \sqrt{8} = 3\sqrt{2}$ is still irrational, but that case is not covered by the mn -criterion ($mn = 16$ is a square) and needs a separate observation. - The proof never needs the sharper fact that $\sqrt{2}$ and $\sqrt{3}$ are individually irrational, nor any theory of field degrees or minimal polynomials; one squaring suffices.

Formal status

The Lean statement above was rebuilt from the Frozen Claim and checked by a fresh Lean 4 / Mathlib toolchain by Hardy's independent FinalVerifier, and its axiom dependencies were audited: the formal theorem is kernel verified. The formal proof follows the exposition step for step — it introduces the hypothetical rational q by unfolding 'Irrational', supplies $(q^2 - 5)/2$ as the rational witness for $\sqrt{6}$, establishes $q^2 = 5 + 2\sqrt{6}$ from 'Real.sq_sqrt' and 'Real.sqrt_mul', and closes against 'Irrational ($\sqrt{6}$)'.

The English exposition in this document is a human-readable rendering of that formal argument. The prose itself — including the hand-written descent argument for $\sqrt{6}$ given above, which is an illustrative substitute for the 'norm_num'-certified step rather than a transcription of it — has not been independently machine-checked; only the Lean

statement and proof carry that guarantee.

Known gaps

- The English exposition has not itself been machine-checked. Only the Lean theorem and its proof term were rebuilt and kernel verified; the prose is a human-readable rendering and could in principle diverge from the formal text in wording or emphasis.
- The self-contained parity/descent argument for the irrationality of $\sqrt{6}$ given in the ‘Preliminary’ section is presented for the reader’s benefit and is not the route taken formally. The formal proof obtains ‘Irrational ($\sqrt{6}$)’ from Mathlib’s ‘norm_num’ irrationality extension, which certifies it by locating 6 strictly between the consecutive squares 2^2 and 3^2 . The two arguments establish the same statement, but only the latter is the one that was verified.
- The generalization mentioned in the Remarks — that $\sqrt{m} + \sqrt{n}$ is irrational whenever mn is not a perfect square — is stated without proof and was not formalized. Only the specific case $m = 2$, $n = 3$ is verified.
- The final rearrangement relies on the standard fact that the coercion $\mathbb{Q} \rightarrow \mathbb{R}$ is a ring homomorphism, so that $(q^2 - 5)/2$ computed in $\mathbb{Q}$ maps to $((q : \mathbb{R})^2 - 5)/2$ in $\mathbb{R}$. This is used silently in the prose; in Lean it is discharged by ‘push_cast’ together with linear arithmetic, and is part of the verified proof, but a reader should note it is an appeal to the coercion’s algebraic compatibility rather than an evident identification.

Exact Lean statement

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theorem irrational_sqrt_two_add_sqrt_three : Irrational (Real.sqrt (2 : ℝ) + Real.sqrt (3
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Verification appendix

Axioms used: propext, Classical.choice, Quot.sound

Frozen Claim SHA-256: 8c21bc2ba346c1a4f8dcc7de768d258522a9b51ba91290759377c2b299750dbc
Lean source SHA-256: f329ece01705b287a80e34ef886918d96d5bb1427378fa3a720720c39a9482c2
Verification SHA-256: 328fd2f404128c4c7dc35ddabc5c79c9aba3c1249f7738c23478f2e435dfe261
Prompt set SHA-256: 777f19c97622765dc491943e1d658d959ad114a7301d9d8654a366a36abf5180
Tectonic bundle SHA-256: 6ffe055852f8faf66c0acbel1a7fb27f87b869a90bad1204f3bf4d9683f597c7c

Run and tool identities

- Run ID: 7676b533-d3dd-4598-bbf0-477a2cce7b13
- Model: claude-opus-5
- Backend: claude
- Runtime SDK: 0.2.127
- Lean: 4.33.1
- Mathlib: 0df444a360eaa60ab8c11dca51a86af692955474
- Tectonic: 0.16.9
- Tectonic bundle: <https://data1.fullyjustified.net/tlextras-2022.0r0.tar>

Lean kernel verification, faithfulness (approved by the user and read back by an independent model before proof search), heuristic informal review, and document compilation are independent statuses. This trusted-development MVP does not sandbox generated Lean or TeX.